POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6023D | Semester VI | Theory | 4 Credits

Computer Integrated Manufacturing

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6023D as Computer Integrated Manufacturing in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user's approved public-source approach, this handbook is transparently based on the NPTEL IIT Kanpur Computer Integrated Manufacturing course and its course preview. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. CNC execution, robot/cell operation, PLC wiring, production data access and machine modification require authorised procedures, verified programs, access control, guarding, isolation and competent supervision.

Verified source note: Verified sources support CIM, computer graphics/geometric modelling, CNC and CAM, group technology/CAPP, FMS/robotics/PLC/AIDC, CAQC/CMM, rapid manufacturing, material handling, simulation, data handling and current digital-manufacturing context. The four-module order is a learning sequence, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Computer Integrated Manufacturing studies how digital information links product design, planning, machining, material movement, quality, control and enterprise decisions. It develops a systems view of CAD/CAM/CNC, process planning, flexible automation, data capture, quality and simulation while respecting the safety and governance boundaries of real manufacturing systems.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Manufacturing processes, engineering graphics/CAD, CNC awareness, basic computer concepts, measurement/quality basics and safe workshop practice.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain CIM concepts, computer roles and information integration across manufacturing.
2. Explain CAD/CAM, geometric modelling, CNC machining, tooling and programme-data concepts.
3. Explain group technology, CAPP, FMS, robotics, PLC, AIDC and material-handling integration.
4. Explain computer-aided quality, rapid manufacturing, simulation, data management and responsible digital-manufacturing practice.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain CIM architecture, information flow and the role of computers in manufacturing systems.	Understand
CO2	Explain CAD/CAM, geometric modelling, CNC machining and part-programming concepts.	Understand
CO3	Explain integration through group technology, CAPP, FMS, robotics, PLC, AIDC and material handling.	Understand
CO4	Explain computer-aided quality, simulation, rapid manufacturing, data governance and continuous improvement in CIM.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	CIM Concepts, Architecture and Information Flow	CIM definition and benefits; design-to-production information flow; computer, database and communication roles; system boundaries, integration needs and human oversight.	Approx. 14 hours	CO1	Module 1
2	CAD/CAM, CNC and Digital Manufacturing Data	Computer graphics and geometric modelling; CAD/CAM link; CNC machine/tooling concepts; part-programming logic, verification and controlled execution.	Approx. 16 hours	CO2	Module 2
3	Planning, Flexible Automation and Shop-Floor Integration	Group technology, CAPP, FMS, robots, PLC, automatic identification/data capture and material handling as coordinated CIM elements.	Approx. 16 hours	CO3	Module 3
4	Quality, Simulation, Rapid Manufacturing and Improvement	CAQC and CMM awareness; data-driven inspection and traceability; rapid/additive manufacturing connection; plant simulation, current digital manufacturing and continuous improvement.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

CIM Concepts, Architecture and Information Flow

CIM definition and benefits; design-to-production information flow; computer, database and communication roles; system boundaries, integration needs and human oversight.

CO1 · Approx. 14 hours

CAD/CAM, CNC and Digital Manufacturing Data

Computer graphics and geometric modelling; CAD/CAM link; CNC machine/tooling concepts; part-programming logic, verification and controlled execution.

CO2 · Approx. 16 hours

Planning, Flexible Automation and Shop-Floor Integration

Group technology, CAPP, FMS, robots, PLC, automatic identification/data capture and material handling as coordinated CIM elements.

CO3 · Approx. 16 hours

Quality, Simulation, Rapid Manufacturing and Improvement

CAQC and CMM awareness; data-driven inspection and traceability; rapid/additive manufacturing connection; plant simulation, current digital manufacturing and continuous improvement.

CO4 · Approx. 14 hours

MODULE 1

CIM Concepts, Architecture and Information Flow

CIM definition and benefits; design-to-production information flow; computer, database and communication roles; system boundaries, integration needs and human oversight.

Approx. 14 hours | CO1 | Understand · Apply

Learning goals

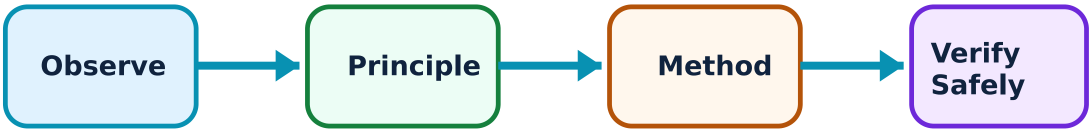
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for CIM Concepts, Architecture and Information Flow: identify the system, state its principle, follow the correct method and verify the result safely.

1. What CIM integrates−

CIM connects digital product information with planning, production, handling, quality and management functions. Its intended benefits include better control, flexibility, speed, traceability and coordinated decision-making, provided the information and physical systems are integrated reliably.

Study and application method

Draw a high-level information-flow map from customer/design requirement through CAD, planning, production, inspection and feedback. Label data owner, input/output, decision point and physical process at each stage.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a high-level information-flow map from customer/design requirement through CAD, planning, production, inspection and feedback. Label data owner, input/output, decision point and physical process at each stage.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Digital integration does not remove the need for engineering review, quality control or safe work. Poor data, weak interfaces or unauthorised changes can propagate quickly through a system.

2. Manufacturing data and system boundaries−

CIM relies on defined product, process, resource, schedule and quality information. Databases, interfaces and standards support information sharing, while access control, version control, traceability and backup protect decision integrity.

Study and application method

For one manufacturing record, identify required fields, source, approver, revision status, user, retention need and risk if the record is incorrect or unavailable.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For one manufacturing record, identify required fields, source, approver, revision status, user, retention need and risk if the record is incorrect or unavailable.

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Common mistake / precaution: Production designs, programs and quality data can be sensitive. Do not copy, alter or share real organisational data without authorisation and governance controls.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

CAD/CAM, CNC and Digital Manufacturing Data

Computer graphics and geometric modelling; CAD/CAM link; CNC machine/tooling concepts; part-programming logic, verification and controlled execution.

Approx. 16 hours | **CO2** | **Understand · Apply**

Learning goals

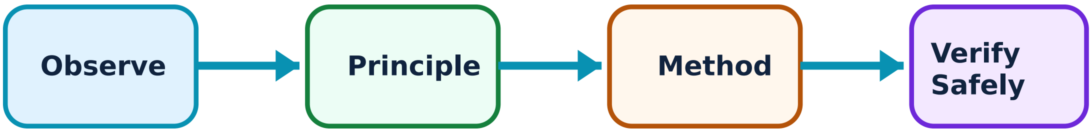
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for CAD/CAM, CNC and Digital Manufacturing Data: identify the system, state its principle, follow the correct method and verify the result safely.

1. Geometric modelling and CAD/CAM connection−

CAD represents part geometry and associated design information; CAM uses validated geometry and process data to help generate manufacturing operations/tool paths. The link reduces manual translation but still needs review of geometry, tolerances, tools, fixtures and process limits.

Study and application method

Trace a simple part from drawing/model to manufacturing plan, identifying design intent, reference datums, geometry check, tooling/fixture decision, tool path, simulation and inspection requirement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace a simple part from drawing/model to manufacturing plan, identifying design intent, reference datums, geometry check, tooling/fixture decision, tool path, simulation and inspection requirement.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A model that displays correctly may still contain tolerancing, geometry, orientation or version issues. Do not rely on visual appearance alone before manufacturing.

2. CNC machining and part-program concepts−

CNC combines programmed motion with machine control. Programming expresses coordinate systems, tool motions, feeds/speeds, operations and auxiliary functions within a defined controller/machine environment. Tooling, setup and verification strongly affect outcome.

Study and application method

Interpret an instructor-supplied, non-live program conceptually: identify coordinate/reference information, motion intent, tooling assumptions, safety checks and required dry-run/simulation evidence.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Interpret an instructor-supplied, non-live program conceptually: identify coordinate/reference information, motion intent, tooling assumptions, safety checks and required dry-run/simulation evidence.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never run, edit or transfer a program to a machine unless authorised. Verified program, correct work offset, tool/fixture condition, guarding and supervised prove-out are essential.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Planning, Flexible Automation and Shop-Floor Integration

Group technology, CAPP, FMS, robots, PLC, automatic identification/data capture and material handling as coordinated CIM elements.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

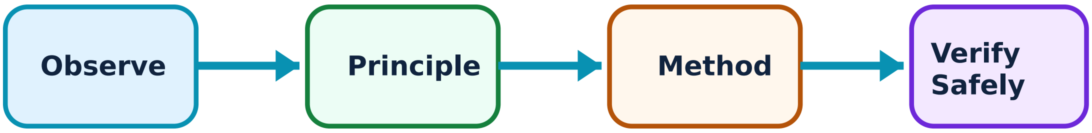
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Planning, Flexible Automation and Shop-Floor Integration: identify the system, state its principle, follow the correct method and verify the result safely.

1. Group technology and computer-aided process planning−

Group technology identifies part families with similar design/manufacturing characteristics; CAPP supports consistent planning by linking part features, resources and operations. Both aim to reduce planning effort and improve standardisation when classification data are maintained.

Study and application method

Classify example parts by selected features/operations, explain the rationale for a part family and outline a provisional process plan with assumptions and missing data.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Classify example parts by selected features/operations, explain the rationale for a part family and outline a provisional process plan with assumptions and missing data.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Part-family and planning decisions can affect tooling, capacity and quality. Use verified classification standards and current resource information before operational use.

2. FMS, robotics, PLC, AIDC and handling−

Flexible systems combine programmable machines, material handling, control, tools and information to handle product variation. Robots/PLCs coordinate motion and logic; AIDC supports identification/traceability; material handling moves and presents work safely and predictably.

Study and application method

Map a small automated cell: part entry, identification, operation, handling, controller, quality check, exception path and human intervention point. Identify an interface that needs validation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map a small automated cell: part entry, identification, operation, handling, controller, quality check, exception path and human intervention point. Identify an interface that needs validation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Integrated cells have hazards from unexpected motion, energy, stored parts and software states. Do not bypass guards/interlocks or change PLC/robot logic without authorised safety validation.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Quality, Simulation, Rapid Manufacturing and Improvement

CAQC and CMM awareness; data-driven inspection and traceability; rapid/additive manufacturing connection; plant simulation, current digital manufacturing and continuous improvement.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

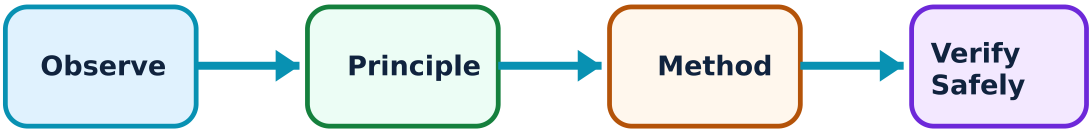
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Quality, Simulation, Rapid Manufacturing and Improvement: identify the system, state its principle, follow the correct method and verify the result safely.

1. Computer-aided quality and measurement−

Computer-aided quality can connect inspection planning, measurement data, CMM results, statistical review and corrective feedback. The purpose is evidence-based conformance and learning, not merely collecting more data.

Study and application method

Create a quality-data path from requirement to measurement plan, calibration status, result record, acceptance decision, nonconformance response and feedback to design/process.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a quality-data path from requirement to measurement plan, calibration status, result record, acceptance decision, nonconformance response and feedback to design/process.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Measurement output is valid only when the method, calibration, environment, datum strategy and uncertainty are appropriate. Do not treat automated data as self-validating.

2. Simulation, rapid manufacturing and digital improvement−

Simulation can test layouts, material flow, resource utilisation and scenarios before physical changes. Rapid/additive manufacturing can support prototypes, tooling or selected production strategies. Improvement requires valid data, controlled trials and human review of outcomes.

Study and application method

Frame a simulation/improvement question with scope, inputs, assumptions, performance measures, validation data, decision owner and condition for escalating or stopping the study.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Frame a simulation/improvement question with scope, inputs, assumptions, performance measures, validation data, decision owner and condition for escalating or stopping the study.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Simulation outputs reflect their inputs and assumptions. Avoid using a model to justify a major change without validation, safety review and appropriate stakeholder approval.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: CIM Concepts, Architecture and Information Flow

Use the concept flow to revise observation, principle, method and verification.

Module 2: CAD/CAM, CNC and Digital Manufacturing Data

Use the concept flow to revise observation, principle, method and verification.

Module 3: Planning, Flexible Automation and Shop-Floor Integration

Use the concept flow to revise observation, principle, method and verification.

Module 4: Quality, Simulation, Rapid Manufacturing and Improvement

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Create a CIM information-flow map for a simple component from design to inspection feedback.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Trace a CAD-to-CAM-to-CNC workflow and identify the mandatory review/verification points.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a part-family and provisional CAPP worksheet for a small supplied set of components.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Map an automated manufacturing cell showing PLC/robot/AIDC/material-handling interfaces and safety boundaries.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Design a CAQC data path and a simulation-study brief for an improvement case.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — CIM Concepts, Architecture and Information Flow

Explain What CIM integrates and state one correct method, application or precaution. −

CIM connects digital product information with planning, production, handling, quality and management functions. Its intended benefits include better control, flexibility, speed, traceability and coordinated decision-making, provided the information and physical systems are integrated reliably.

Answer framework: Draw a high-level information-flow map from customer/design requirement through CAD, planning, production, inspection and feedback. Label data owner, input/output, decision point and physical process at each stage.

Important point: Digital integration does not remove the need for engineering review, quality control or safe work. Poor data, weak interfaces or unauthorised changes can propagate quickly through a system.

Explain Manufacturing data and system boundaries and state one correct method, application or precaution.−

CIM relies on defined product, process, resource, schedule and quality information. Databases, interfaces and standards support information sharing, while access control, version control, traceability and backup protect decision integrity.

Answer framework: For one manufacturing record, identify required fields, source, approver, revision status, user, retention need and risk if the record is incorrect or unavailable.

Important point: Production designs, programs and quality data can be sensitive. Do not copy, alter or share real organisational data without authorisation and governance controls.

Module 2 — CAD/CAM, CNC and Digital Manufacturing Data

Explain Geometric modelling and CAD/CAM connection and state one correct method, application or precaution.−

CAD represents part geometry and associated design information; CAM uses validated geometry and process data to help generate manufacturing operations/tool paths. The link reduces manual translation but still needs review of geometry, tolerances, tools, fixtures and process limits.

Answer framework: Trace a simple part from drawing/model to manufacturing plan, identifying design intent, reference datums, geometry check, tooling/fixture decision, tool path, simulation and inspection requirement.

Important point: A model that displays correctly may still contain tolerancing, geometry, orientation or version issues. Do not rely on visual appearance alone before manufacturing.

Explain CNC machining and part-program concepts and state one correct method, application or precaution.−

CNC combines programmed motion with machine control. Programming expresses coordinate systems, tool motions, feeds/speeds, operations and auxiliary functions within a defined controller/machine environment. Tooling, setup and verification strongly affect outcome.

Answer framework: Interpret an instructor-supplied, non-live program conceptually: identify coordinate/reference information, motion intent, tooling assumptions, safety checks and required dry-run/simulation evidence.

Important point: Never run, edit or transfer a program to a machine unless authorised. Verified program, correct work offset, tool/fixture condition, guarding and supervised prove-out are essential.

Module 3 — Planning, Flexible Automation and Shop-Floor Integration

Explain Group technology and computer-aided process planning and state one correct method, application or precaution.−

Group technology identifies part families with similar design/manufacturing characteristics; CAPP supports consistent planning by linking part features, resources and operations. Both aim to reduce planning effort and improve standardisation when classification data are maintained.

Answer framework: Classify example parts by selected features/operations, explain the rationale for a part family and outline a provisional process plan with assumptions and missing data.

Important point: Part-family and planning decisions can affect tooling, capacity and quality. Use verified classification standards and current resource information before operational use.

Explain FMS, robotics, PLC, AIDC and handling and state one correct method, application or precaution.−

Flexible systems combine programmable machines, material handling, control, tools and information to handle product variation. Robots/PLCs coordinate motion and logic; AIDC supports identification/traceability; material handling moves and presents work safely and predictably.

Answer framework: Map a small automated cell: part entry, identification, operation, handling, controller, quality check, exception path and human intervention point. Identify an interface that needs validation.

Important point: Integrated cells have hazards from unexpected motion, energy, stored parts and software states. Do not bypass guards/interlocks or change PLC/robot logic without authorised safety validation.

Module 4 — Quality, Simulation, Rapid Manufacturing and Improvement

Explain Computer-aided quality and measurement and state one correct method, application or precaution.−

Computer-aided quality can connect inspection planning, measurement data, CMM results, statistical review and corrective feedback. The purpose is evidence-based conformance and learning, not merely collecting more data.

Answer framework: Create a quality-data path from requirement to measurement plan, calibration status, result record, acceptance decision, nonconformance response and feedback to design/process.

Important point: Measurement output is valid only when the method, calibration, environment, datum strategy and uncertainty are appropriate. Do not treat automated data as self-validating.

Explain Simulation, rapid manufacturing and digital improvement and state one correct method, application or precaution.−

Simulation can test layouts, material flow, resource utilisation and scenarios before physical changes. Rapid/additive manufacturing can support prototypes, tooling or selected production strategies. Improvement requires valid data, controlled trials and human review of outcomes.

Answer framework: Frame a simulation/improvement question with scope, inputs, assumptions, performance measures, validation data, decision owner and condition for escalating or stopping the study.

Important point: Simulation outputs reflect their inputs and assumptions. Avoid using a model to justify a major change without validation, safety review and appropriate stakeholder approval.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: CIM Concepts, Architecture and Information Flow.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: CAD/CAM, CNC and Digital Manufacturing Data.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Planning, Flexible Automation and Shop-Floor Integration.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Quality, Simulation, Rapid Manufacturing and Improvement.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: CIM definition and benefits; design-to-production information flow; computer, database and communication roles; system boundaries, integration needs and human oversight..
2. Develop a complete answer or practical plan for: Computer graphics and geometric modelling; CAD/CAM link; CNC machine/tooling concepts; part-programming logic, verification and controlled execution..
3. Develop a complete answer or practical plan for: Group technology, CAPP, FMS, robots, PLC, automatic identification/data capture and material handling as coordinated CIM elements..
4. Develop a complete answer or practical plan for: CAQC and CMM awareness; data-driven inspection and traceability; rapid/additive manufacturing connection; plant simulation, current digital manufacturing and continuous improvement..

LAST-MILE REVISION

Quick Revision

Module 1: CIM Concepts, Architecture and Information Flow

CIM definition and benefits; design-to-production information flow; computer, database and communication roles; system boundaries, integration needs and human oversight.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: CAD/CAM, CNC and Digital Manufacturing Data

Computer graphics and geometric modelling; CAD/CAM link; CNC machine/tooling concepts; part-programming logic, verification and controlled execution.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Planning, Flexible Automation and Shop-Floor Integration

Group technology, CAPP, FMS, robots, PLC, automatic identification/data capture and material handling as coordinated CIM elements.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Quality, Simulation, Rapid Manufacturing and Improvement

CAQC and CMM awareness; data-driven inspection and traceability; rapid/additive manufacturing connection; plant simulation, current digital manufacturing and continuous improvement.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
What CIM integrates	CIM connects digital product information with planning, production, handling, quality and management functions.
Manufacturing data and system boundaries	CIM relies on defined product, process, resource, schedule and quality information.
Geometric modelling and CAD/CAM connection	CAD represents part geometry and associated design information; CAM uses validated geometry and process data to help generate manufacturing operations/tool paths.
CNC machining and part-program concepts	CNC combines programmed motion with machine control.
Group technology and computer-aided process planning	Group technology identifies part families with similar design/manufacturing characteristics; CAPP supports consistent planning by linking part features, resources and operations.
FMS, robotics, PLC, AIDC and handling	Flexible systems combine programmable machines, material handling, control, tools and information to handle product variation.
Computer-aided quality and measurement	Computer-aided quality can connect inspection planning, measurement data, CMM results, statistical review and corrective feedback.
Simulation, rapid manufacturing and digital improvement	Simulation can test layouts, material flow, resource utilisation and scenarios before physical changes.

LEARNING RESOURCES

References

Recommended computer-integrated-manufacturing references

1. M. P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing.
2. P. N. Rao, CAD/CAM: Principles and Applications.
3. M. P. Groover and E. W. Zimmers, CAD/CAM: Computer-Aided Design and Manufacturing.
4. S. S. Rao, Engineering Optimization (for structured systems thinking).

Approved public computer-integrated-manufacturing sources

- https://nptel.ac.in/courses/112104289
- https://onlinecourses.nptel.ac.in/noc24_me27/preview

These sources provide the verified study basis while official Course 6023D detail is unavailable; they do not replace CNC/controller manuals, approved programs, enterprise data rules, safety standards or authorised manufacturing procedures.